

Due on tuesday, March 28th, in class

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1. Consider a set of inertial frames which are aligned at the commonly defined zero time, but might be moving in different relative directions.
 - (a) [**1+2 pts**] Write down the Lorentz boost matrices for relative velocity v_1 between frames in the x -direction, and for relative velocity v_2 in the y -direction.
 - (b) [**5 pts**] Find out by matrix multiplication the result of successive application of these two transformations. Consider both possible orderings: applying the x -direction boost first and then the y -direction boost, and vice versa.
This will look cumbersome, so I suggest that you use $c = 1$ units and the notation $\gamma_1 = \gamma(v_1)$, $\gamma_2 = \gamma(v_2)$. Another hint: do you really need to multiply 4×4 matrices?
 - (c) [**2 pts**] Do the two Lorentz boost matrices commute? Compare with Lorentz boosts in the same direction.
 - (d) [**1+2 pts**] Consider any of the matrices resulting from successive application of the two boosts. Is the matrix symmetric?
Remembering from Problem Set 5 that Lorentz boost transformations in arbitrary directions are symmetric matrices, interpret this observation.
2. Relative velocities, longitudinal & transverse.
 - (a) [**2 pts**] A rocket traveling at speed $\frac{1}{2}c$ with respect to frame Σ shoots forward bullets at speed $\frac{3}{4}c$ relative to the rocket. What is the speed of the bullets relative to Σ ?
 - (b) [**4 pts**] Now imagine that the bullets are fired *perpendicular* to the direction of rocket motion, as perceived from the rocket. What is the speed of the bullets relative to Σ ?
 - (c) [**3+2 pts**] An X-ray beam is sent out from the rocket, perpendicular to the direction of rocket motion. Relative to Σ , what are the components of the X-ray beam velocity parallel and perpendicular to the direction of rocket motion? Explain why the speed relative to Σ could have been expected.

3. **[5 pts]** Show that, if the 4×4 transformation matrix Λ acting on 4-vectors (ct, x, y, z) preserves the Minkowski norm $c^2t^2 - x^2 - y^2 - z^2$, then it must satisfy $\Lambda^T g \Lambda = g$. Here g is the metric tensor defined in Problem Set 05.
4. The kinetic energy of motion of a particle is the relativistic total energy minus the rest energy.
- (a) **[3 pts]** A particle has rest mass M and speed v . If $v \ll c$, then show that the kinetic energy due to motion is approximated by the well-known non-relativistic expression for the kinetic energy.
- (b) **[3 pts]** Derive the leading correction to the non-relativistic expression.
- (c) **[3 pts]** What is a particle's speed if its kinetic energy T is equal to its rest energy? What is a particle's speed if its kinetic energy is n times its rest energy?
- (d) **[3 pts]** Express the momentum p of the particle as a function of the mass M and the kinetic energy of motion, which we will call T . Your expression should contain M and T , not v or γ_v .
Hint: use the relation between E and p that does not involve v or γ_v .
5. Spacetime diagrams (plots of ct versus x).
- (a) **[3 pts]** On a spacetime diagram, draw the worldline of a photon starting at the origin, the worldline of an object with speed $c/2$ starting at the same time at the origin, and the worldline of an object with speed $4c$ starting at the same time at the origin. Which worldline is unphysical?
- (b) **[2 pts]** On a spacetime diagram, draw the worldline of a photon starting at the origin, and the worldline of an object with velocity $c/2$ starting at the origin four seconds later.
- (c) **[2 pts]** On a spacetime diagram, depict the situation that a light source at $x = 0$ is turned on for a few seconds, and then turned off again. (Show the worldlines of photons.)
- (d) **[2 pts]** A railgun located at the origin shoots projectiles at time intervals of T , alternately in the positive- x and negative- x directions. The projectiles in the positive x direction have speed $c/10$ and those in the negative x direction have speed $c/2$. Draw the situation on a spacetime diagram, showing the worldlines of a number of successive projectiles.